Computational Stats Project

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1 Fisher Scoring for GLM

Fisher Scoring is an iterative optimization method used to find MLE estimates of a model's parameters. As we discussed in class, it is a variant of the Newton–Raphson method, where the observed Fisher information (i.e. the observed Hessian matrix), present in the update formula, is replaced by its expected version. This method is relevant for likelihood equations of GLMs, as they often do not have explicit solutions, and thus need this kind of optimization methods. Fisher scoring is particularly convenient for GLMs because the Fisher information matrix ultimately has this simple form $I(\beta) = X^\top W X$, so that each iteration reduces to solving a weighted least squares problem.
The actual implementation of the algorithm (as described in the lecture notes) is:
starting from an initial β_0, iterate for $t = 0, 1, 2, \ldots$:
(i) compute $\eta_t = X\beta_t$ (ii)compute the link function $\mu_t = g^{-1}(\eta_t)$ (iii) compute $z_t = \eta_t + (y - \mu_t)\frac{\partial \eta_t}{\partial \mu_t}$
(iv) compute the weights (diagonal matrix W_t) where $w_{i,i} = \frac{1}{\mathrm{Var}(Y_i)}\left(\frac{\partial \eta_{t,i}}{\partial \mu_{t,i}}\right)^{-2}$ (v) update $\beta_{t+1} = (X^\top W_t X)^{-1} X^\top W_t z_t$
Stop when β_t converges (for example, when the difference in the update is below a certain tolerance).

1.1 Bernoulli model with logit link

Since $Y_i \in \{0, 1\}$, we can model it as $Y_i \mid x_i \sim \text{Bernoulli}(p_i)$, and define $\mu_i = E[Y_i] = p_i$. In logistic regression we use the logit link and the linear predictor

$$\eta_i = \log\left(\frac{\mu_i}{1-\mu_i}\right) = x_i^\top \beta, \qquad \Longleftrightarrow \qquad \mu_i = \frac{e^{\eta_i}}{1+e^{\eta_i}}. \qquad \mathrm{Var}(Y_i) = \mu_i(1-\mu_i).$$

Log-likelihood and score Let X be the design matrix with rows $x_i^\top$ and $\eta = X\beta$. The log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log(\mu_i) + (1-y_i)\log(1-\mu_i) \Big] = \sum_{i=1}^{n} \Big[y_i \eta_i - \log(1+e^{\eta_i}) \Big],$$

hence, using $\frac{\partial \eta_i}{\partial \beta} = x_i$ and $\frac{\partial}{\partial \eta_i}\log(1+e^{\eta_i}) = \frac{e^{\eta_i}}{1+e^{\eta_i}} = \mu_i$, we derive the gradient (score function) as

$$\nabla \ell(\beta) = \sum_{i=1}^{n} x_i\,(y_i - \mu_i) = X^\top (y - \mu).$$

Fisher scoring update Differentiate the score once more:

$$\nabla^2 \ell(\beta) = \frac{\partial}{\partial \beta}\big[X^\top (y-\mu)\big] = -X^\top \frac{\partial \mu}{\partial \beta}$$

The chain rule tells us that $\frac{\partial \mu}{\partial \beta} = \frac{\partial \mu}{\partial \eta}\frac{\partial \eta}{\partial \beta}$. Since $\eta = X\beta$ we have $\frac{\partial \eta}{\partial \beta} = X$, and for $\mu_i = \frac{e^{\eta_i}}{1+e^{\eta_i}}$,

$$\frac{\partial \mu_i}{\partial \eta_i} = \mu_i(1-\mu_i) \quad \Rightarrow \quad \frac{\partial \mu}{\partial \eta} = W, \qquad \text{where} \quad W = \mathrm{diag}(w_1, \ldots, w_n), \quad w_i = \mu_i(1-\mu_i),$$

$$\nabla^2 \ell(\beta) = -X^\top W X, \qquad \mathcal{I}(\beta) = -E[\nabla^2 \ell(\beta)] = X^\top W X.$$

Substituting, the Fisher scoring iteration becomes

$$\beta_{t+1} = \beta_t + \mathcal{I}(\beta_t)^{-1}\nabla \ell(\beta_t) = \beta_t + (X^\top W_t X)^{-1} X^\top (y - \mu_t),$$

where $\eta_t = X\beta_t$, $\mu_{t,i} = \frac{e^{\eta_{t,i}}}{1+e^{\eta_{t,i}}}$ and $(W_t)_{ii} = \mu_{t,i}(1-\mu_{t,i})$. After defining z_t as

$$z_{t,i} = \eta_{t,i} + \frac{y_i - \mu_{t,i}}{\mu_{t,i}(1-\mu_{t,i})}, \qquad i = 1, \ldots, n,$$

the update takes the form

$$\beta_{t+1} = (X^\top W_t X)^{-1} X^\top W_t z_t.$$

1.2 Poisson model with log link

Assume $Y_i \in N_0$, $Y_i \mid x_i \sim \text{Poisson}(\mu_i)$, with $\mu_i = E[Y_i]$. In Poisson regression we use the canonical log link and the linear predictor

$$\eta_i = \log(\mu_i) = x_i^\top \beta, \qquad \Longleftrightarrow \qquad \mu_i = e^{\eta_i}. \qquad \mathrm{Var}(Y_i) = \mu_i.$$

Log-likelihood and score Let X be the design matrix with rows $x_i^\top$ and $\eta = X\beta$. The log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \Big[y_i \log(\mu_i) - \mu_i - \log(y_i!) \Big] = \sum_{i=1}^{n} \Big[y_i \eta_i - e^{\eta_i} - \log(y_i!) \Big].$$

Using $\frac{\partial \eta_i}{\partial \beta} = x_i$ and $\frac{\partial}{\partial \eta_i} e^{\eta_i} = \mu_i$,

$$\nabla \ell(\beta) = \sum_{i=1}^{n} x_i \,(y_i - \mu_i) = X^\top (y - \mu).$$

Fisher scoring update As we did before, we can differentiate the score to obtain the Hessian: $-X^\top \frac{\partial \mu}{\partial \beta}$
The chain rule tells us that $\frac{\partial \mu}{\partial \beta} = \frac{\partial \mu}{\partial \eta} \frac{\partial \eta}{\partial \beta}$. Since $\eta = X\beta$ we have $\frac{\partial \eta}{\partial \beta} = X$, and for $\mu_i = e^{\eta_i}$,

$$\frac{\partial \mu_i}{\partial \eta_i} = \mu_i \quad \Rightarrow \quad \frac{\partial \mu}{\partial \eta} = W \qquad \text{where} \quad W = \mathrm{diag}(w_1, \dots, w_n), \quad w_i = \mu_i = e^{\eta_i},$$

thus

$$\nabla^2 \ell(\beta) = -X^\top W X, \qquad \mathcal{I}(\beta) = -E[\nabla^2 \ell(\beta)] = X^\top W X.$$

The Fisher scoring iteration is

$$\beta_{t+1} = \beta_t + \mathcal{I}(\beta_t)^{-1} \nabla \ell(\beta_t) = \beta_t + (X^\top W_t X)^{-1} X^\top (y - \mu_t),$$

where $\eta_t = X\beta_t$, $\mu_{t,i} = e^{\eta_{t,i}}$ and $(W)_{ii} = \mu_{t,i}$. After defining z_t as

$$z_{t,i} = \eta_{t,i} + \frac{y_i - \mu_{t,i}}{\mu_{t,i}}, \qquad i = 1, \dots, n,$$

the update takes the form

$$\beta_{t+1} = (X^\top W_t X)^{-1} X^\top W_t z_t.$$

1.3 Proof: Fisher scoring coincides with Newton's method for canonical links

For an exponential family distribution

$$f(y_i; \theta_i, \phi) = \exp\left\{ \frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right\}, \qquad \mu_i = E(Y_i) = b'(\theta_i).$$

With a canonical link, we have $\theta_i = \eta_i = x_i^\top \beta$. Hence the log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \left[\frac{y_i \eta_i - b(\eta_i)}{a(\phi)} + c(y_i, \phi) \right].$$

Differentiate using the fact that $\partial \eta_i / \partial \beta = x_i$:

$$\nabla_\beta \ell(\beta) = \frac{1}{a(\phi)} \sum_{i=1}^{n} x_i \big(y_i - b'(\eta_i) \big) = \frac{1}{a(\phi)} X^\top (y - \mu),$$

and

$$H(\beta) = \nabla_\beta^2 \ell(\beta) = -\frac{1}{a(\phi)} \sum_{i=1}^{n} b''(\eta_i)\, x_i x_i^\top = -\frac{1}{a(\phi)} X^\top \mathrm{diag}\big(b''(\eta_i)\big) X.$$

What we can notice here is that $H(\beta)$ does not depend on the random y_i, therefore $I(\beta) := E[-H(\beta)] = -H(\beta)$. Which means that Newton ($\beta_{new} = \beta - H(\beta)^{-1} \nabla \ell(\beta)$) and Fisher scoring ($\beta_{new} = \beta + I(\beta)^{-1} \nabla \ell(\beta)$) are identical since $I(\beta) = -H(\beta)$.

1.4 Discussion on non-canonical links

First of all, with a non-canonical link, $\theta_i \neq \eta_i$. This means that the observed Hessian contains an additional term which depends on the data, thus making the expected value of the Fisher information $I(\beta) \neq -H(\beta)$. Thus Newton and Fisher scoring do not necessarily coincide anymore.
In addition, for a general GLM, Fisher scoring can be written (as derived in the notes on GLM) as: $\beta_{t+1} = (X^\top W_t X)^{-1} X^\top W_t z_t$ with $z_t = \eta_t + (y - \mu_t) \left(\frac{d\eta_t}{d\mu_t} \right)$ and diagonal weights $W_t = \frac{\left(\frac{d\mu_t}{d\eta_t} \right)^2}{Var(Y_t)}$ where $Var(Y_i) = \phi V(\mu_i)$.
For a canonical link one has

$$\frac{d\eta_i}{d\mu_i} = \frac{1}{V(\mu_i)} \qquad \Longrightarrow \qquad w_i = \frac{1}{V(\mu_i) \left(\frac{1}{V(\mu_i)} \right)^2} = V(\mu_i),$$

e.g. $w_i = \mu_i(1 - \mu_i)$ for logit-Bernoulli and $w_i = \mu_i$ for log-Poisson. For non-canonical links this cancellation does not occur, thus yielding more complicated weights.

2 Metropolis–Hastings for Generalized Linear Models

In the Bayesian framework, the regression coefficients are treated as random variables and inference is based on their posterior distribution. For logistic and Poisson regression models, the posterior distribution has no closed-form expression due to the non-Gaussian likelihood and the nonlinear link function, making Markov Chain Monte Carlo methods necessary. MCMC algorithms allow the posterior distribution to be explored through simulation by generating samples from a Markov chain with the desired stationary distribution. In this project, a Random Walk Metropolis–Hastings algorithm is used, as it provides a flexible sampling scheme that only requires evaluation of the posterior density and does not rely on closed-form full conditional distributions.These methodological choices are applied to both the Bayesian Logistic regression and the Bayesian Poisson regression.

2.1 Posterior distribution and log-posterior formulation

The posterior distribution of the regression coefficients is given by

$$\pi(\beta \mid y, X) \propto p(y \mid X, \beta)\, p(\beta),$$

where $p(y \mid X, \beta)$ denotes the likelihood function and $p(\beta)$ the prior distribution on the regression coefficients. The proportionality sign indicates that the posterior density is defined up to a normalizing constant, this constant ensures that the posterior integrates to one but does not depend on β and therefore cancels out in Metropolis–Hastings acceptance ratios.
For computational reasons, computations are performed on the log scale. The log-posterior is thus defined as

$$\log \pi(\beta \mid y, X) = \log p(y \mid X, \beta) + \log p(\beta) + C,$$

where C is a constant independent of β and irrelevant for inference.

2.2 Prior specification

All covariates are standardized prior to the analysis. We assume a weakly informative Gaussian prior on the regression coefficients:

$$\beta \sim \mathcal{N}(0, \tau^2 I),$$

with $\tau = 5$.
In the absence of strong prior information, this prior reflects the belief that covariates have a limited effect on the outcome, and therefore that the regression coefficients are likely to be close to zero. At the same time, the chosen variance allows the posterior to explore a wide range of plausible coefficient values while discouraging implausibly large effects. Since all covariates are standardized, the same prior variance can be meaningfully applied to all coefficients, and the choice $\tau = 5$ llows the coefficients to vary over a wide but plausible range, without encouraging extreme values.

2.3 Metropolis–Hastings algorithm

Starting from an initial value $\beta^{(0)}$, the Metropolis–Hastings algorithm generates a Markov chain $\{\beta^{(t)}\}_{t=1}^{T}$ by iterating the same update rule for a fixed number of iterations T. At each iteration t, a candidate value $\beta^\star$ is proposed according to a random-walk mechanism:

$$\beta^\star = \beta^{(t)} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \Sigma),$$

where Σ denotes the proposal covariance matrix.
Two different MCMC runs are performed, corresponding to alternative choices of initialization and proposal covariance.
In the first run, the chain is initialized at $\beta^{(0)} = 0$, and an isotropic random-walk proposal is used with covariance matrix

$$\Sigma = \sigma^2 I,$$

where σ^2 is a scalar tuning parameter controlling the overall step size of the random walk.
In the second run, the chain is initialized at the frequentist maximum likelihood estimate $\hat{\beta}_{\mathrm{MLE}}$, and the proposal covariance matrix is based on the estimated covariance matrix obtained from the Fisher scoring algorithm. Specifically, the proposal covariance is defined as

$$\Sigma = s\, V_{\mathrm{Fisher}},$$

where V_{Fisher} captures the local curvature of the log-likelihood, while the scalar factor $s > 0$ is a parameter that controls the global step size of the proposal.

2.4 Acceptance probability

Given the current state $\beta^{(t)}$ and a proposed value $\beta^\star$ drawn from a proposal distribution $q(\beta^\star \mid \beta^{(t)})$, the Metropolis–Hastings acceptance probability is

$$\alpha(\beta^{(t)}, \beta^\star) = \min\left\{1,\ \frac{\pi(\beta^\star)\, q(\beta^{(t)} \mid \beta^\star)}{\pi(\beta^{(t)})\, q(\beta^\star \mid \beta^{(t)})}\right\},$$

where $\pi(\cdot)$ denotes the posterior density.
Using a symmetric random-walk proposal,

$$q(\beta^\star \mid \beta^{(t)}) = q(\beta^{(t)} \mid \beta^\star), \qquad \alpha(\beta^{(t)}, \beta^\star) = \min\left\{1,\ \frac{\pi(\beta^\star)}{\pi(\beta^{(t)})}\right\}.$$

For computational convenience, the acceptance probability is evaluated on the log scale by defining

$$\log \alpha = \log \pi(\beta^\star) - \log \pi(\beta^{(t)}),$$

and the proposed value is accepted if $\log U \leq \log \alpha$, where $U \sim \mathrm{Uniform}(0, 1)$.

2.5 Tuning and convergence assessment

The σ^2 and s parameters of the two versions (isotropic and fisher-informed) proposal covariance matrix are tuned to achieve an acceptance rate between 0.2 and 0.4, which is commonly recommended for random-walk Metropolis–Hastings algorithms, as it provides a balance between exploration of the parameter space and acceptance probability. Convergence is assessed by comparing posterior summaries for increasing values of the chain length T. Results stabilize for $T = 20{,}000$ iterations, in the sense that posterior means and credible intervals show negligible changes for larger values of T. Finally, a burn-in period corresponding to the first 30% of the chain is discarded.

3 Results

3.1 Intepretation of coefficients and covariates

Since coefficients estimates look quite consistent across the different models, for space reasons, we report here only the ones from the Fisher Scoring algorithm (others are available in the appendix or in the notebook).

term	$\hat{\beta}$	se	p-value
reports	-2.356	0.190	< 0.001
active	0.834	0.119	< 0.001
income	0.383	0.109	< 0.001

(a) Credit Card (logistic).

term	$\hat{\beta}$	se	p-value
procedure	0.564	0.072	< 0.001
type	0.094	0.065	< 0.001
gender	-0.049	0.065	< 0.001

(b) Azcabgptca (Poisson).

Table 1: Frequentist estimates for the two GLMs (standard errors and Wald p-values).

All covariates were standardized prior to estimation to place them on a common scale. For both datasets, the ranking of the most influential covariates is stable across the different model specifications.
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For the Credit Card dataset, the most influential covariates are reports, active and income.
The first one exhibits a negative effect, indicating that a higher number of negative credit reports reduces the probability of credit card acceptance. While, active and income have a positive impact, applicants with active credit accounts and higher income levels are more likely to be granted a credit card.
For the Azcabgptca dataset, the three most influential covariates are procedure, type and gender.
The first one has a positive effect on the expected length of hospital stay, indicating that certain medical procedures are associated with significantly longer hospitalization periods. The covariate type also increases the expected number of days spent in hospital, since the emergency cases are typically associated with more severe clinical conditions and require longer hospitalization. While gender shows a negative effect. Since gender $= 1$ corresponds to male patients, this implies that male patients are associated with a slightly shorter expected length of hospital stay compared to female patients.

3.2 Diagnostic for convergence of MCMC

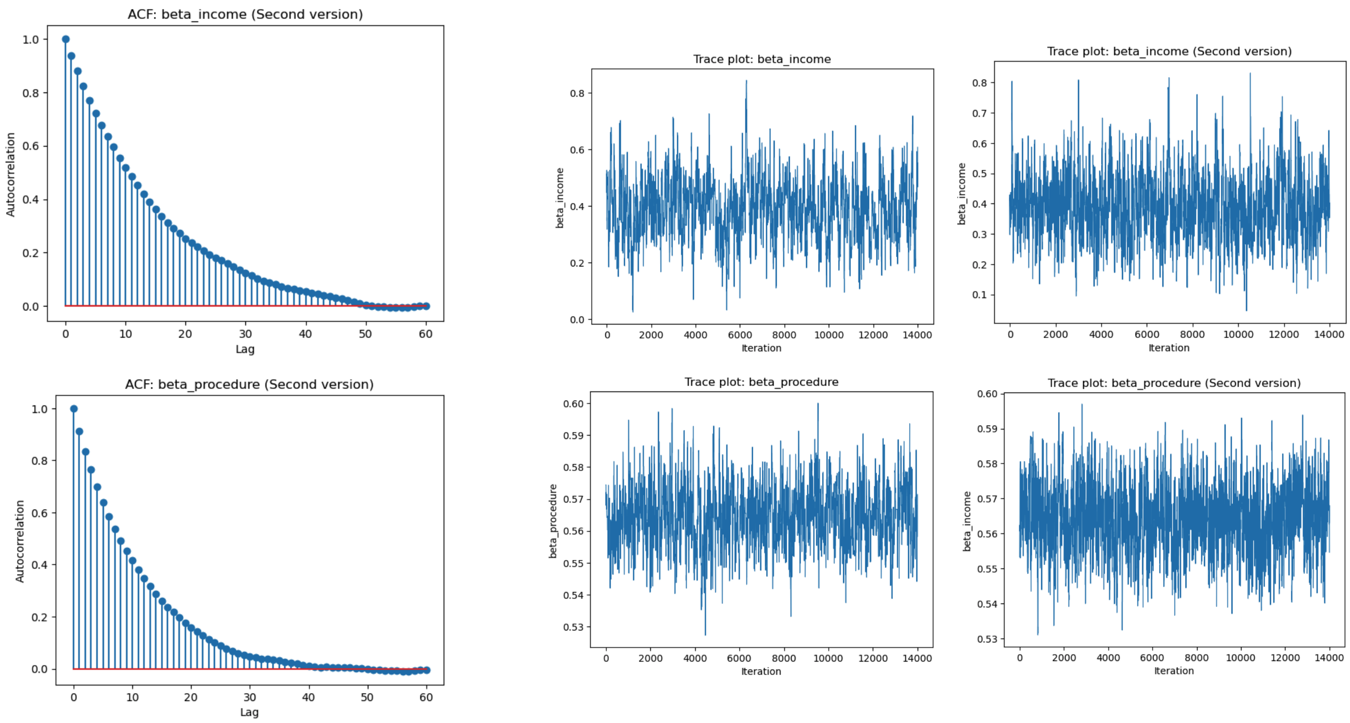

Figure 1: MCMC diagnostics. Left: autocorrelation functions (ACF). Right: traceplots for selected parameters.

The trace plots for beta_income and beta_procedure both appear stable over all the iterations, fluctuating around a roughly constant mean with no evident long-term trend, drift or structural breaks. This behavior is consistent with the chains having reached a stationary distribution and suggests satisfactory convergence from a qualitative perspective. The autocorrelation function (ACF) plots provide a clear comparison between the two implementations (standard vs second version). In particular, for the credit card dataset the ACF second version decays towards zero at substantially smaller lags, indicating lower serial dependence between successive draws. This indicates that a better-tuned configuration, in terms of starting point and proposal distribution, can significantly enhance MCMC performance.

4 Appendix

term	$\hat{\beta}$	**se**	**p-value**
reports	-2.356	0.190	< 0.001
active	0.834	0.119	< 0.001
income	0.383	0.109	< 0.001

Table 2: Credit Card dataset: frequentist estimates (logistic regression).

term	**mean**	**sd**	**95% CI**
reports	-2.433	0.194	$(-2.818, -2.057)$
active	0.866	0.121	(0.626, 1.107)
income	0.393	0.108	(0.186, 0.614)

Table 3: Credit Card dataset: posterior summaries.

term	**mean**	**sd**	**95% CI**
reports	-2.386	0.190	$(-2.773, -2.025)$
active	0.849	0.118	(0.627, 1.089)
income	0.386	0.108	(0.183, 0.600)

Table 4: Credit Card dataset: posterior summaries using frequentist initial values.

term	$\hat{\beta}$	**se**	**p-value**
procedure	0.564	0.072	< 0.001
type	0.094	0.065	< 0.001
gender	-0.049	0.065	< 0.001

Table 5: Azcabgptca dataset: frequentist estimates (Poisson regression).

term	**mean**	**sd**	**95% CI**
procedure	0.566	0.010	(0.547, 0.585)
type	0.095	0.009	(0.078, 0.111)
gender	-0.049	0.008	$(-0.065, -0.033)$

Table 6: Azcabgptca dataset: posterior summaries.

term	**mean**	**sd**	**95% CI**
procedure	0.564	0.010	(0.545, 0.583)
type	0.095	0.008	(0.077, 0.111)
gender	-0.048	0.009	$(-0.066, -0.032)$

Table 7: Azcabgptca dataset: posterior summaries using frequentist initial values.